THE HEFFERNAN GROUP
P.O. Box 1180
Surrey Hills North VIC 3127
Phone 03 9836 5021
info@theheffernangroup.com.au
www.theheffernangroup.com.au

SPECIALIST MATHEMATICS TRIAL EXAMINATION 1 SOLUTIONS 2024

Question 1 (3 marks)

Evaluate $\int_{\frac{1}{2}}^{1} \frac{\log_e(2x)}{x^2}\,dx$ using integration by parts.

The integration by parts formula is given on the formula sheet: $\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$.

Let $u = \log_e(2x)$ and let $\frac{dv}{dx} = \frac{1}{x^2}$.

$\therefore \frac{du}{dx} = \frac{1}{x}$ and $v = -\frac{1}{x}$.

$$\therefore \int_{\frac{1}{2}}^{1} \frac{\log_e(2x)}{x^2}\,dx = \left[-\frac{1}{x}\log_e(2x)\right]_{\frac{1}{2}}^{1} - \int_{\frac{1}{2}}^{1} -\frac{1}{x}\times\frac{1}{x}\,dx$$

(1 mark)

$$= \left[-\frac{1}{x}\log_e(2x)\right]_{\frac{1}{2}}^{1} - \left[\frac{1}{x}\right]_{\frac{1}{2}}^{1}$$

(1 mark)

$$= (-\log_e(2) - 0) - (1 - 2)$$

$$= 1 - \log_e(2)$$

(1 mark)

Question 2 (3 marks)

$8\sin^4(x)\cot^4(x) + \cos(2x) + 12\sin^2(x) = 8$

$8\sin^4(x)\dfrac{\cos^4(x)}{\sin^4(x)} + \cos(2x) + 12\sin^2(x) = 8$

$8\cos^4(x) + (2\cos^2(x) - 1) + 12(1 - \cos^2(x)) = 8$, $\sin(x) \neq 0$

(1 mark – simplify cot^4(x) or re-write cos(2x))

$8\cos^4(x) - 10\cos^2(x) + 3 = 0$

Let $a = \cos^2(x)$

$8a^2 - 10a + 3 = 0$

$(4a - 3)(2a - 1) = 0$

$a = \dfrac{3}{4}, \dfrac{1}{2}$

(1 mark – recognise quadratic = 0)

$\therefore \cos^2(x) = \dfrac{3}{4}, \dfrac{1}{2}$

$\cos(x) = \pm\dfrac{\sqrt{3}}{2}, \pm\dfrac{1}{\sqrt{2}}$

Given $x \in \left(\dfrac{\pi}{2}, \pi\right)$ (quadrant 2)

$x = \dfrac{3\pi}{4}, \dfrac{5\pi}{6}$

(1 mark)

Question 3 (4 marks)

Let I be the normal random variable for the volume (mL) of ice-cream dispensed.
Let T be the normal random variable for the volume (mL) of chocolate topping dispensed.

$\mathrm{E}(I) = 125 \qquad \mathrm{E}(T) = 20$
$\mathrm{Var}(I) = 16 \qquad \mathrm{Var}(T) = 9$

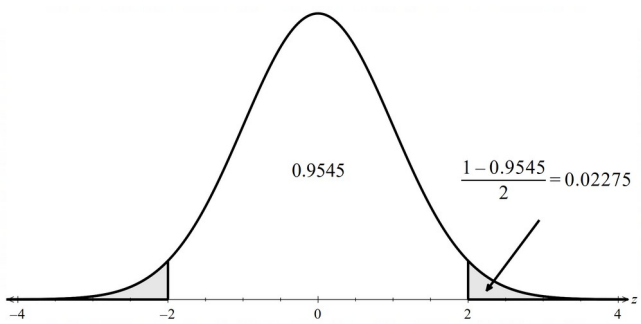
a. Let $V = I + T$, where V is the random variable for the volume (mL) of ice-cream and chocolate topping dispensed.

$\mathrm{E}(V) = 125 + 20 = 145$
$\mathrm{Var}(V) = 16 + 9 = 25$ since I and T are independent random variables
$\therefore \mathrm{sd}(I + T) = \sqrt{25} = 5$

(1 mark – either E(*I* + *T*) or Var(*I* + *T*) correct)

$\therefore \Pr(V > 155)$

$= \Pr(Z > 2)$ since $Z = \dfrac{V - \mu_V}{\sigma_V} = \dfrac{155 - 145}{5} = 2$
$= 0.02275$
$= 0.023$ (correct to 3 decimal places) **(1 mark)**



b. Cost = $\dfrac{1}{1000}(11I + 15T)$

(note: multiply by $\dfrac{1}{1000}$ to account for conversion from mL to L)

$\mathrm{E}(\text{Cost}) = \dfrac{1}{1000}(11\mathrm{E}(I) + 15\mathrm{E}(T))$ **(1 mark)**
$= \dfrac{1}{1000}(11 \times 125 + 15 \times 20)$
$= \dfrac{1}{1000}(1675)$
$= \$1.68$ (nearest cent) **(1 mark)**

Question 4 (3 marks)

Let $P(n)$ be the proposition that $7+9+11+....+(2n+1)=n^2+2n-8$ for all integers $n \geq 3$.

<u>Step 1</u> Show $P(3)$ is true.

$7=3^2+2\times 3-8$

$7=7$ which is true. **(1 mark)**

<u>Step 2</u> Assume that $P(k)$ is true for $k \in Z$, $k \geq 3$.

$7+9+11+....+(2k+1)=k^2+2k-8$

<u>Step 3</u> Prove that $P(k)$ true implies that $P(k+1)$ is true.

$$\begin{aligned}
&7+9+11+....+(2k+1)+(2(k+1)+1)\\
&=7+9+11+....+(2k+1)+(2k+3)\\
&=P(k)+(2k+3)\\
&=k^2+2k-8+(2k+3)\\
&=k^2+4k-5\\
&=k^2+(2k+2k)+1+2-8\\
&=k^2+2k+1+2k+2-8\\
&=(k+1)^2+2(k+1)-8\\
&=P(k+1)
\end{aligned}$$

(1 mark)

It follows that $P(k)$ true implies that $P(k+1)$ is true.
Using the principle of mathematical induction, it therefore follows that $P(n)$ is true for all integers $n \geq 3$. **(1 mark)**

Question 5 (3 marks)

a. The scalar resolute of $\underset{\sim}{b}$ in the direction of $\underset{\sim}{a}$ is $\dfrac{\underset{\sim}{b}\cdot a}{|\underset{\sim}{a}|}$.

$$\therefore \frac{\underset{\sim}{b}\cdot a}{|\underset{\sim}{a}|} = \frac{7}{\sqrt{14}}$$

$|\underset{\sim}{a}| = \sqrt{3^2 + (-1)^2 + 2^2} = \sqrt{14}$ and $\underset{\sim}{a}\cdot b = 3\times 1 - m + 2\times 1 = 5 - m$

$$\therefore \frac{7}{\sqrt{14}} = \frac{5-m}{\sqrt{14}}$$

$7 = 5 - m$
$m = -2$ **(1 mark)**

b. Let $\underset{\sim}{u}$ be the vector resolute of $\underset{\sim}{b}$ in the direction of $\underset{\sim}{a}$.

$$\underset{\sim}{u} = \frac{\underset{\sim}{b}\cdot a}{\underset{\sim}{a}\cdot a} a = \frac{7}{\sqrt{14}\sqrt{14}}\underset{\sim}{a} = \frac{1}{2}\underset{\sim}{a} = \frac{1}{2}(3\underset{\sim}{i} - \underset{\sim}{j} + 2k)$$ **(1 mark)**

Let $\underset{\sim}{v}$ be the vector resolute of $\underset{\sim}{b}$ that is perpendicular to $\underset{\sim}{a}$.

Then $\underset{\sim}{b} = u + v \Rightarrow \underset{\sim}{v} = b - u$

$\therefore$ vector resolute of $\underset{\sim}{b}$ that is perpendicular to $\underset{\sim}{a}$ is $\underset{\sim}{v} = b - \frac{1}{2}a$

$$\begin{aligned}\underset{\sim}{b} - \frac{1}{2}a &= (\underset{\sim}{i} + m\underset{\sim}{j} + k) - \frac{1}{2}(3i - j + 2k)\\ &= (\underset{\sim}{i} - 2\underset{\sim}{j} + k) - \frac{1}{2}(3i - j + 2k)\\ &= -\frac{1}{2}\underset{\sim}{i} - \frac{3}{2}\underset{\sim}{j}\end{aligned}$$ **(1 mark)**

Alternatively, you can use $|\underset{\sim}{a}| = \sqrt{14}$ and scalar resolute of $\underset{\sim}{b}$ in the direction of $\underset{\sim}{a} = \dfrac{7}{\sqrt{14}}$ to determine that $\underset{\sim}{u} = \frac{1}{2}a$.

Question 6 (6 marks)

a. $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 2$

Using implicit differentiation,

$$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}} \times \frac{dy}{dx} = 0$$

(1 mark)

<u>Method 1</u> – Rearrange for $\frac{dy}{dx}$ first

$$\frac{dy}{dx} = \frac{-\frac{2}{3}x^{-\frac{1}{3}}}{\frac{2}{3}y^{-\frac{1}{3}}}$$

$$= -\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}}$$

Sub in the point $(1,1)$,

$$\frac{dy}{dx} = -1$$

(1 mark)

<u>Method 2</u> – Substitute in the point $(1,1)$ into $\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}} \times \frac{dy}{dx} = 0$.

$$\frac{2}{3} \times 1 + \frac{2}{3} \times 1 \times \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -1$$

b. Using formula sheet, Surface Area $=\int_0^{\pi} 2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\ dt$

$x=2^{\frac{3}{2}}\cos^3(t)$ $\qquad$ $y=2^{\frac{3}{2}}\sin^3(t)$

$\frac{dx}{dt}=-3\times 2^{\frac{3}{2}}\cos^2(t)\sin(t)$ $\qquad$ $\frac{dy}{dt}=3\times 2^{\frac{3}{2}}\sin^2(t)\cos(t)$

(1 mark – either derivative correct)

$$SA=2\pi\int_0^{\pi}2^{\frac{3}{2}}\sin^3(t)\sqrt{(-3\times 2^{\frac{3}{2}}\cos^2(t)\sin(t))^2+(3\times 2^{\frac{3}{2}}\sin^2(t)\cos(t))^2}\ dt$$

$$=2\pi\times 2^{\frac{3}{2}}\int_0^{\pi}\sin^3(t)\sqrt{9\times 2^3\cos^4(t)\sin^2(t)+9\times 2^3\sin^4(t)\cos^2(t)}\ dt$$

$$=2\pi\times 2^{\frac{3}{2}}\int_0^{\pi}\sin^3(t)\sqrt{(9\times 2^3\cos^2(t)\sin^2(t))(\sin^2(t)+\cos^2(t))}\ dt$$

(1 mark – get to point of simplifying $\cos^2(t)+\sin^2(t)$)

$$=2\pi\times 2^{\frac{3}{2}}\int_0^{\pi}\sin^3(t)\sqrt{(9\times 2^3\cos^2(t)\sin^2(t)}\ dt$$

$$=2\pi\times 2^{\frac{3}{2}}\int_0^{\pi}\sin^3(t)\times 3\times 2^{\frac{3}{2}}\,|\cos(t)\sin(t)|\ dt$$

since $\sqrt{\cos^2(t)\sin^2(t)}=|\cos(t)\sin(t)|$

$$=6\pi\times 2^3\int_0^{\pi}\sin^3(t)\,|\cos(t)\sin(t)|\ dt$$

$$=48\pi\int_0^{\pi}\sin^3(t)\,|\cos(t)\sin(t)|\ dt$$

Given symmetry of graph, it is easier to use domain $0\le t\le\frac{\pi}{2}$ as this gives $\cos(t)\ge 0$ and $\sin(t)\ge 0$ (therefore $|\cos(t)\sin(t)|=\cos(t)\sin(t)$) and multiply by 2.

$$\therefore SA=2\times 48\pi\int_0^{\frac{\pi}{2}}\sin^4(t)\cos(t)\ dt$$

(1 mark)

$$=96\pi\int_0^{\frac{\pi}{2}}\sin^4(t)\cos(t)\ dt$$

Let $u=\sin(t)$

$\frac{du}{dt}=\cos(t)$

$t=0$, $u=0$

$$=96\pi\int_0^{\frac{\pi}{2}}u^4\frac{du}{dt}dt$$

$t=\frac{\pi}{2}$, $u=1$

$$=96\pi\int_0^1 u^4\ du$$

$$=96\pi\left[\frac{u^5}{5}\right]_0^1$$

$$=\frac{96\pi}{5}\ \text{units}^2$$ **(1 mark)**

Question 7 (3 marks)

$2\overline{z}^2+2z+\text{Re}(z)=-5$

Let $z=x+yi$, $x,y\in R$.

$2(x-yi)^2+2(x+yi)+x+5=0$

$2x^2-4xyi-2y^2+2x+2yi+x+5=0$ **(1 mark)**

Equating real parts: $2x^2-2y^2+3x+5=0$ …… (1)

Equating imaginary parts: $-4xy+2y=0$ ……. (2)

From (2), $y(-4x+2)=0$

$y=0$ or $x=\frac{1}{2}$

- Substitute $y=0$ into (1)

  $2x^2+3x+5=0$

  Check discriminant, $\Delta=3^2-4\times2\times5=-31<0$

  $\therefore$ no real solution for x therefore y = 0 is rejected.

  (1 mark – determine only one solution for *x*)

- Substitute $x=\frac{1}{2}$ into (1)

$$2\left(\frac{1}{2}\right)^2-2y^2+3\left(\frac{1}{2}\right)+5=0$$

$$-2y^2=-7$$

$$y^2=\frac{7}{2}$$

$$y=\pm\sqrt{\frac{7}{2}}=\pm\frac{\sqrt{7}}{\sqrt{2}}=\pm\frac{\sqrt{14}}{2}$$

$$\therefore\ z=\frac{1}{2}-\frac{\sqrt{14}}{2}i,\quad z=\frac{1}{2}+\frac{\sqrt{14}}{2}i$$ **(1 mark)**

Question 8 (4 marks)

$$\frac{dv}{dt} = \frac{2v}{1+t^2}$$

$$\therefore \int \frac{1}{v}\, dv = \int \frac{2}{1+t^2}\, dt$$

(1 mark – attempt to separate variables)

$$\log_e |v| = 2\tan^{-1}(t) + c$$

(1 mark)

<u>Method 1 – rearrange the constant</u>

$$|v| = e^{2\tan^{-1}(t)+c}$$
$$= e^c e^{2\tan^{-1}(t)}$$
$$\therefore v = \pm e^c e^{2\tan^{-1}(t)}$$
$$= Ae^{2\tan^{-1}(t)} \quad \text{where } A = \pm e^c \in R \setminus \{0\}$$

Substitute in $v(0) = e$:

$$e = Ae^{2\tan^{-1}(0)} = A$$

(1 mark – attempt to find *A*)

Therefore $v = e \times e^{2\tan^{-1}(t)}$

$$v = e^{2\tan^{-1}(t)+1}$$

(1 mark)

<u>Method 2 – find the +*c* first</u>

$$\log_e |v| = 2\tan^{-1}(t) + c$$

Given $v(0) = e$,

$$1 = 2\tan^{-1}(0) + c$$
$$c = 1$$

(1 mark – attempt to find +*c*)

$$\log_e |v| = 2\tan^{-1}(t) + 1$$
$$v = \pm e^{2\tan^{-1}(t)+1}$$

Using the given condition of $v(0) = e$, leads us to reject the negative answer.

$$v = e^{2\tan^{-1}(t)+1}$$

(1 mark)

Question 9 (4 marks)

a. By inspection, $\underset{\sim}{d}_1 = \sqrt{2}\underset{\sim}{i} + a\underset{\sim}{j} + 3k$ is a vector in the direction of the line $\underset{\sim}{r}_1$ and $\underset{\sim}{d}_2 = \sqrt{2}\underset{\sim}{i} + \underset{\sim}{j} + k$ is a vector in the direction of the line $\underset{\sim}{r}_2$.

By definition, the angle between two lines is equal to the angle between these two vectors and is found using the dot product.

Using the dot product, $\underset{\sim}{d}_1 \cdot \underset{\sim}{d}_2 = |d_1||d_2|\cos(\theta)$,

$$\therefore\ 5 + a = \sqrt{11 + a^2} \times \sqrt{4} \times \cos(60^\circ)$$ **(1 mark)**

$$5 + a = \sqrt{11 + a^2}$$

$$25 + 10a + a^2 = 11 + a^2$$

$$14 = -10a$$

$$a = \frac{-7}{5}$$ **(1 mark)**

Since the 'squaring' process can introduce extraneous solutions, it should be checked that $a = \frac{-7}{5}$ satisfies the original equation (which it does).

b. A normal to the plane is a vector normal to the direction of each line, that is, normal to the vectors $\underset{\sim}{d}_1 = \sqrt{2}\underset{\sim}{i} - \frac{7}{5}\underset{\sim}{j} + 3k$ and $\underset{\sim}{d}_2 = \sqrt{2}\underset{\sim}{i} + \underset{\sim}{j} + k$.

The cross product is used to determine this normal vector and hence, the equation of the plane.

Using the formula sheet for vectors $\underset{\sim}{r}_1 = x_1 i + y_1 \underset{\sim}{j} + z_1 k$ and $\underset{\sim}{r}_2 = x_2 i + y_2 \underset{\sim}{j} + z_2 k$,
$\underset{\sim}{r}_1 \times \underset{\sim}{r}_2 = (y_1 z_2 - y_2 z_1) i + (x_2 z_1 - x_1 z_2) \underset{\sim}{j} + (x_1 y_2 - x_2 y_1) k$.

$$\therefore\ \underset{\sim}{d}_1 \times d_2 = \left(\frac{-7}{5} \times 1 - 1 \times 3\right) \underset{\sim}{i} + (\sqrt{2} \times 3 - \sqrt{2} \times 1) \underset{\sim}{j} + \left(\sqrt{2} \times 1 - \sqrt{2} \times -\frac{7}{5}\right) k$$
$$= -\frac{22}{5} \underset{\sim}{i} + 2\sqrt{2} \underset{\sim}{j} + \frac{12\sqrt{2}}{5} \underset{\sim}{k}$$

$\therefore\ \frac{-22}{5}x + 2\sqrt{2}y + \frac{12\sqrt{2}}{5}z = k$, $k \in R$ is the equation of the plane. **(1 mark)**

Substitute in a point on the plane e.g. $(5, \sqrt{2}, -5\sqrt{2})$ (which was found when $\mu = 0$ for line $\underset{\sim}{r}_2(\mu) = 5i + \sqrt{2}\underset{\sim}{j} - 5\sqrt{2}k + \mu(\sqrt{2}i + j + k)$).

(Another point that could be used as an example is $\left(10, -\sqrt{2}, \frac{5\sqrt{2}}{4}\right)$ found when $\lambda = 0$ for $\underset{\sim}{r}_1(\lambda) = 10i - \sqrt{2}\underset{\sim}{j} + \frac{5\sqrt{2}}{4}k + \lambda(\sqrt{2}i + aj + 3k)$).

$$\frac{-22}{5} \times 5 + 2\sqrt{2} \times \sqrt{2} + \frac{12\sqrt{2}}{5} \times -5\sqrt{2} = k$$

$$-22 + 4 - 24 = k$$

$$k = -42$$

$\therefore\ \frac{-22}{5}x + 2\sqrt{2}y + \frac{12\sqrt{2}}{5}z = -42$ is the equation of the plane. **(1 mark)**

(alternatives include $\frac{22}{5}x - 2\sqrt{2}y - \frac{12\sqrt{2}}{5}z = 42$ or $22x - 10\sqrt{2}y - 12\sqrt{2}z = 210$)

Question 10 (7 marks)

a.

$$f(x) = \frac{2x^2 - 4}{(x-2)^2}$$

$$\therefore f'(x) = \frac{(x-2)^2(4x) - (2x^2-4)(2(x-2))}{(x-2)^4}$$

$$= \frac{(x-2)(4x(x-2) - 2(2x^2-4))}{(x-2)^4}$$

(1 mark – attempt quotient rule and simplification)

$$= \frac{4x^2 - 8x - 4x^2 + 8}{(x-2)^3}$$

$$= \frac{-8x+8}{(x-2)^3}$$

Stationary point occurs when $f'(x) = 0$

$$\therefore -8x + 8 = 0$$
$$x = 1$$

$$f(1) = \frac{2-4}{(-1)^2} = -2$$

$\therefore$ stationary point is at $(1, -2)$. **(1 mark – show that)**

b.

$$f''(x) = \frac{(x-2)^3(-8) - (-8x+8)(3(x-2)^2)}{(x-2)^6}$$

$$= \frac{(x-2)^2(-8(x-2) - 3(-8x+8))}{(x-2)^6}$$

$$= \frac{16x-8}{(x-2)^4}$$

A condition for a point of inflection to occur is that $f''(x) = 0$

$$\frac{16x-8}{(x-2)^4} = 0$$

$$\therefore\ 16x - 8 = 0$$

$$x = \frac{1}{2}$$

(1 mark – show that)

A second condition for a point of inflection to occur is that there must be a change in concavity (that is, a change in sign of $f''(x)$).

x	0	$\frac{1}{2}$	1
$f''(x)$	$= \frac{-8}{16} < 0$	0	$= \frac{8}{1} > 0$

(1 mark – show sign change)

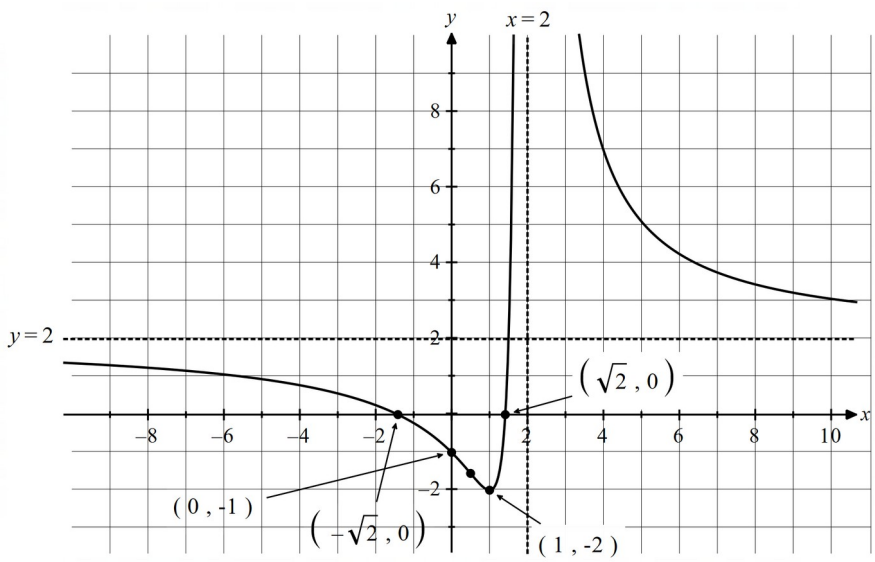
c.

- Stationary point given at $(1, -2)$
- y-intercept: $f(0) = -1$
- x-intercepts: $f(x) = 0$

  $2x^2 - 4 = 0$

  $x = \pm\sqrt{2}$
- asymptotes: $(x-2)^2 = 0$ $\therefore x = 2$ vertical asymptote

  $\dfrac{2x^2 - 4}{(x-2)^2} = 2 + \dfrac{8x - 12}{(x-2)^2}$ $\therefore y = 2$ horizontal asymptote



(1 mark – two asymptotes with their equations)
(1 mark – correct shape including point of inflection shape indicated)
(1 mark – correct *y*-intercept, *x*-intercepts and stationary point)